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ORIGINAL RESEARCH

Suitable habitats shifting toward human-dominated landscapes of Asian elephants in China

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Abstract
Tropical forests are undergoing land use change in many regions of the world, with an increasingly prominent trend of landscape fragmentation. Nevertheless, the Asian elephant (*Elephas maximus*) population in tropical forests of China has continued to increase, and their distribution range is expanding to human-modified landscapes, resulting in increased severe human-elephant conflicts. Little is known about the habitat adaptation strategies of Asian elephants in human-dominated habitats. Here, we applied maximum entropy modeling to extensive occurrence data (21,011 independent records) of Asian elephants in southwest China to explore temporal changes in their distribution under climate change. We also identified critical factors shaping their current distribution and habitat selection strategies concerning human-induced land-use changes at multiple spatial scales. We found that the area of suitable habitats for Asian elephants will slightly increase in the future (7,025 km^2) compared with the present climatic scenario (6,468 km^2). At a finer spatial scale, topographic variables such as elevation (74%) and slope (11%) are the dominant factors shaping the habitat selection of Asian elephants. However, at a larger spatial scale, anthropogenic variables are the main factors associated with the habitat preference of Asian elephants. Most suitable and sub-suitable habitats (65%) of Asian elephants overlap with human-dominated areas. Our findings suggest that the effects of human activity on Asian elephants are scale-dependent and demonstrate that the suitable habitats of Asian elephants in China are shifting to human-dominated regions.

Keywords Asian elephant · Habitat · Human-dominated regions · Distribution

Qiupeng Yu and Zhechang Hu contributed equally to this work and should be considered co-first authors.

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Introduction

From June to August 2021, the northward movement of a herd of Asian elephants from their original home range in Xishuangbanna to Kunming in far-southwest China garnered global attention. Jiang et al. (2021c); Campos-Arceiz et al. (2022) attributed this movement as a dispersal attempt, which is a typical behavior observed among animals (Nunes et al. 1996; Gosselink et al. 2010; Loe et al. 2010; Debeffe et al. 2012). Dispersal involves individuals leaving their original habitat to establish new habitats or join other population (Lidicker 1975; Stenseth and Lidicker 1992; Ronce 2007; Stieha et al. 2014). Range expansions by dispersal or recolonization occur frequently and widely for Asian elephants in many countries (Baskaran et al. 2011; Gubbi et al. 2014; Neupane et al. 2020; Sivasubramanian and Ramakrishnan 2021). In China, the Asian elephants' range has been expanding from the Mengyang Nature Reserve in Xishuangbanna National Nature Reserve during the early 1990s, subsequently progressing northwards (Yang et al. 1987b; Xiong 1996; Wu et al. 1999; Yu et al. 2015).

The expansion of Asian elephants' range from reserves is driven by multiple factors, such as an increase in vegetation coverage, habitat fragmentation, and population growth (Bowler & Benton 2005; Koirala 2016; Neupane et al. 2020; Talukdar et al. 2020; Taher et al. 2021). In China, the original habitats of Asian elephants within nature reserves have been effectively protected and managed by local nature reserve management bureaus under the Regulations of the State Council with minimal human activity interference since the 1950s (Liu et al. 2020a). Consequently, canopy cover has continuously expanded, while forest understory food sources and accessibility for elephants has reduced (Chen et al. 2006b; Liu et al. 2017b; Campos-Arceiz et al. 2021). On the other hand, the extensive cultivated land outside the reserves may provide supplementary food for elephants and gradually attract them to human-dominated regions (Linkie et al. 2007; Meijaard & Sheil 2013; Dai 2017). These factors have facilitated the elephant dispersal from relatively intact forests within reserves to human-dominated regions with agroforestry ecotones (Huang et al. 2019; de la Torre et al. 2021; Sivasubramanian and Ramakrishnan 2021).

Climate change is widely recognized as a significant driver of shifts in the distribution patterns of wildlife populations (Root et al. 2003; Cohen et al. 2018; Dai et al. 2019; Bai et al. 2022), resulting in certain animals shifting niches towards higher altitudes and latitudes passively (Hickling et al. 2005; Moritz et al. 2008; Chen et al. 2011a; Li et al. 2015b). The direct climate change impacts such as alterations to temperature regimes and water resource availability (Rahman 2009; Ye et al. 2013) and; indirect effects such as changes in vegetation and foraging opportunities (Zubair et al. 2010; Kanagaraj et al. 2019) drive species distribution changes. Recent studies have shown that climate change could expand the potential ranges of some species distribution (Kriticos et al. 2003; Peng et al. 2002; Wuethrich 2000). Relatedly, Wang et al. (2021a) attributed that extreme dry heat caused a dramatic degradation of herbaceous plants and shrubs, consequently triggering the northward migration of Asian elephants in China during 2020–2021.

The progressive dispersal of Asian elephants from reserves as a consequence of habitat fragmentation and loss has led to an increasing overlap between their habitats and human-dominated regions, escalating the frequency of human-elephant conflicts (de la Torre et al. 2021; Ekanayaka et al. 2011; Sukumar 1990; Wilson et al. 2015b). Over the past decade, there has been a surge in human-elephant conflicts, causing numerous injuries or fatalities and extensive property damage (Zhang et al. 2022b). In China, the range of the Asian elephant population consisting of 293 individuals is confined to nine counties comprising three prefectures (Xishuangbanna, Pu'er, and Lincang) in Yunnan Province (Yunnan Provincial Forestry and Grassland Administration, unpublished data). The Asian elephant population in China continues to grow and disperse into human-dominated regions (Huang et al. 2019; Tang et al. 2020; Wang et al. 2021c; Yu et al. 2015). Given the current state of Asian elephant populations, it is imperative to improve the understanding of their dispersal patterns and habitat preferences, and their habitat selection strategy and range change process, especially in human-dominated regions. This information is essential for effective management and conservation efforts. Therefore, our study aimed to reconstruct the historical dispersal process of Asian elephants, focusing on elucidating their dispersal dynamics. We also employed maximum entropy modelling and monitored data to investigate the factors influencing Asian elephants' habitat selection and distribution. We proposed three main research hypotheses:

1) The temporal dynamics of climate may influence species habitat distribution (García-Valdés et al. 2015; Bai et al. 2022), thus, if the Asian elephants' habitat remains stable over time under climate change, it could suggest their capacity to adapt to current climatic shifts.
2) Increased human-elephant interactions have resulted in more significant anthropogenic disturbances within human-dominated elephant habitats (Wilson et al. 2015b; Singh et al. 2023), and the impact of variables on species' habitat selection varies across different spatial scales (Wilson et al. 2013a; Chen et al. 2022c). Therefore, we argue that the significance of human activities on Asian elephant habitat selection differs at various spatial scales.
3) The fragmentation and loss of the natural Asian elephant habitats have led to a reduction in the availability of natural food resources (Sitompul et al. 2013; Koirala 2016; Campos-Arceiz et al. 2021), while agricultural expansion has significantly increased supplementary food sources in the form of artificial crops, attracting Asian elephants towards human-dominated regions (Linkie et al. 2007; Meijaard & Sheil 2013; Chaiyarat et al. 2022). Hence, we hypothesize that suitable habitats for Asian elephants have shifted towards human-dominated regions.

Materials and methods

Study area

The 293 Asian elephants in Yunnan, China, were divided into four populations (Yunnan Provincial Forestry and Grassland Department, Fig. 1). Our study focuses on the Pu'er-Banna population, which is the largest in China, accounting for 64.5% of the total population and originally inhabiting Mengyang Reserve. Over the past three decades, the Pu'er-Banna population frequently traveled between Pu'er and Xishuangbanna and surrounding areas. Currently, the population is distributed in five counties, including Jinghong (Xishuangbanna), Simao, Jiangcheng, Ning'er, and Jinggu (Pu'er) (He 2013; Yu et al. 2015; Zhu et al.

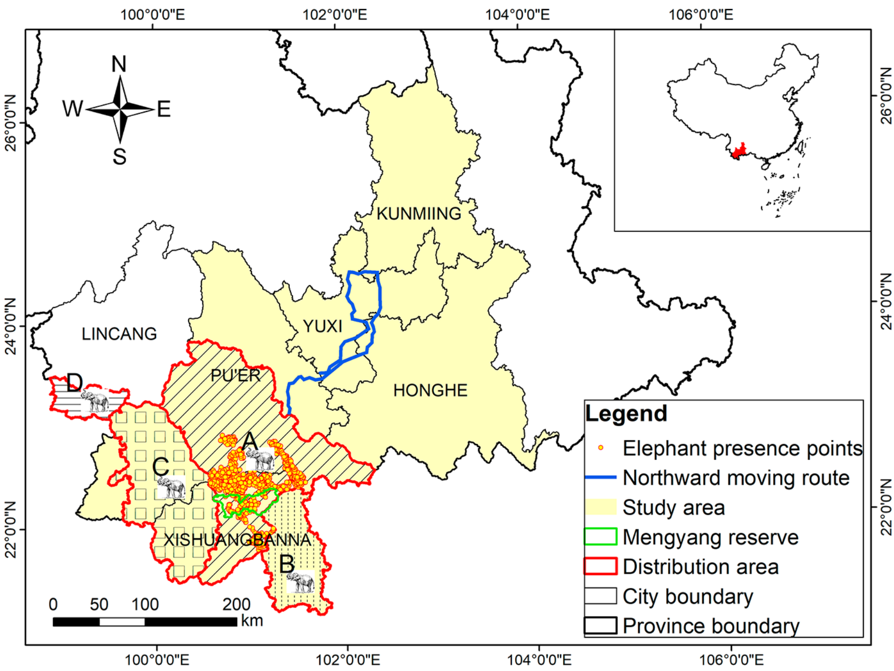

Fig. 1 Study area with presence points of the Pu'er-Banna Asian elephant population and the northward moving route of the Asian elephant herd. The four Asian elephant icons depict the distribution of Asian elephant populations: (**A**) Pu'er-Banna population, (**B**) Mengla-Shangyong population, (**C**) Lancang-Menghai population, (**D**) Cangyuan population (Yunnan Provincial Forestry and Grassland Administration, unpublished data)

2019). Our study area encompassed these five prefectures, extending over an area approximately 131,701 km^2, including regions traversed by the northward-moving herd during 2020–2021 (Fig. 1). The region ranges in elevation from 76 to 4,248 m above sea level and is characterized by tropical and subtropical climates with distinct dry and wet seasons. Subtropical monsoon evergreen-leaved forests dominate the vegetation type in these areas (Jiang 1980a). Approximately 1.6 million individuals are resident in these five prefectures (People's Government of Pu'er City 2022; People's Government of Xishuangbanna Dai Autonomous Prefectural 2021).

Analysis of dispersal history

We obtained historical occurrence records of Asian elephants from the Pu'er Forestry and Grassland Bureau and combined them with data from Yu et al. (2015). Utilizing these records, we investigated the dispersal process of elephants in Pu'er during the 1990s, 2000s, 2010s, and 2020s. Based on the movement corridor width of Asian elephants (Lin et al. 2008; Osborn and Parker 2003; Ravisankar et al. 2019) and previous observations of their movement patterns, we aimed to comprehend these animals' habitat utilization shifts during their dispersal process. Furthermore, we assessed whether there was a tendency for Asian elephants to disperse into human-dominated regions by examining changes in cultivated land within the 5 km buffer zone along the Asian elephant dispersal route.

Habitat analysis

Presence points

We collected 21,011 presence points of Asian elephants between January 2020 and December 2022 (Fig. 1). The data was acquired through active monitoring of Asian elephants by local rangers equipped with unmanned aerial vehicles (UAVs) in collaboration with the local forestry and grassland department. One to two presence points were recorded daily for each individual or herd. The presence points were supplemented with data from 170 infrared camera trapping sites in Mengyang Reserve. A maximum daily movement distance of approximately 10 km was determined based on our previous observations of Asian elephants' movement within the study area. Hence, we utilized three spatial scales of 1 × 1, 5 × 5, and 10 × 10 km grids to evaluate the significance of variables for habitat selection by Asian elephants. We utilized the 1 × 1 km grid with the highest accuracy to facilitated the analysis of the suitability and distribution of Asian elephant habitat. A random selection approach was employed to filter out the presence points within 1 × 1, 5 × 5, and 10 × 10 km grids to mitigate the impact of spatial autocorrelation induced by the proximity of presence points. After spatial filtering, we retained 1,950, 214, and 74 presence points at scales of 1 km, 5 km, and 10 km, respectively.

Environmental and human variables

We obtained 19 bioclimatic data from the WorldClim database versions 1.4 and 2.1 to represent climatic variables for past and present periods (Fick and Hijmans 2017). For future climatic scenarios, we utilized bioclimatic data from the 2041–2060 period based on the general circulation models (GCMs) of CCSM4 of the fifth Assessment Report of the International Panel of Climate Change. The scenarios described by the representative concentration pathway (RCP) 8.5, which represents a worst-case climate change scenario, with continuously rising global temperature, averaging an increase of 4°C by 2100 (Moss et al. 2008; Schwalm et al. 2020; IPCC 2023). To analyze temporal changes in Asian elephant habitat distribution, we employed bioclimatic data of three periods with a spatial resolution of 30″ (~ 1 km). Further details are provided in Appendix Table 1.

To assess the habitat selection and distribution of Asian elephants under human activity impacts across multiple spatial scales, we selected 16 environmental and anthropogenic variables that have been previously identified as influential factors in Asian elephant habitat utilization (Jiang 2019b; Neupane et al. 2020; Wadey 2018). The environmental variables included elevation, slope, aspect, roughness, patch density, normalized difference vegetation index (NDVI), vegetation, river density, and distance to rivers. The anthropogenic variables included human modification, land use, nighttime lights, road density, distance to roads, settlement density, and distance to settlements. Detailed sources and processing methods are in Appendix Tables 2, 3 and 4.

To avoid the effect of collinearity among variables, we retained uncorrelated variables (Spearman correlation absolute values < 0.8) based on the ecological significance (Yackulic et al. 2013). We excluded variables with a contribution of less than 1% (Li et al. 2020c; Yang et al. 2022c). Consequently, we screened 19 bioclimatic variables for the present period, while bio2, bio3, bio5, bio12, bio14, and bio17 were retained for all three periods. The environmental and anthropogenic variables selected at multiple spatial scales summarized in Table 2. The data were resampled to resolutions of 1×1, 5×5, and 10×10 km and projected to WGS_1984_UTM in ArcGIS 10.6.

MaxEnt modeling

The distribution of Asian elephants was modeled using the maximum entropy algorithm (MaxEnt) (Alamgir et al. 2015; Htet et al. 2021; Wang et al. 2018b). We assessed the contribution of variables to estimate their significance in shaping Asian elephant habitat selection, thereby reflecting their habitat preference. The MaxEnt model was used to derive a habitat suitability index (HSI), which predicts the suitability of an area for Asian elephants based on a combination of environmental variables known to influence their occurrence. We developed a climate model and a habitat model. In the climate model, we incorporated only climate variables as explanatory variables to predict the effects of climate change on the habitat suitability of Asian elephants in the temporal dimension (past, present, and future). In the habitat model, we included fine-scale (1 km) environmental and anthropogenic variables underlying the spatial occurrence of Asian elephants, with particular emphasis on the effects of human-induced land-use change on the species' distribution expansion.

We utilized "Maxent Model v3.3.1" and "ENMTools" to establish parameters for modeling the habitat suitability of Asian elephants: linear/quadratic/product = 0.050, categorical = 0.250, threshold = 1.000, hinge = 0.500 (Mudereri et al. 2020). In the MaxEnt setup, we represent the background environment by setting a maximum of 10,000 background points. To ensure model stability, we employed the cross-validation method to select 25% of the Asian elephant presence points randomly as a test set while assigning the remaining 75% to the training set with 20 repetitions. The model was iterated 5,000 times to allow sufficient convergence time while keeping other parameters at their default values (Phillips and Dudík 2008; Elith et al. 2011; Merow et al. 2013). Variable contributions and response curves were generated to analyze the variables' relative importance and impact on prediction results (Li et al. 2019a; Yang et al. 2020a). The prediction results were evaluated using the area under the curve (AUC) from the receiver operating characteristic curve (ROC) (Phillips et al. 2004). The closer AUC values approach 1 with higher model accuracy (Phillips et al. 2004). The difference between the AUC of the training set and the test set can be used for checking overfitting; minor differences suggest less overfitting (Warren and Seifert 2011). The MaxEnt model outputs the average HSI in logistic regression, which was used to classify the Asian elephant habitat into suitable, sub-suitable, and unsuitable categories using the natural breakpoint method.

Analysis of habitat overlap

The extent of overlap between the suitable habitat of Asian elephants and cultivated land can indicate their reliance on crops. To derive this overlap distribution, we focused on the relationship between Asian elephant habitats and human-dominated regions, which often lead to heightened levels of human-elephant conflict (La Due et al. 2021). Human-dominated regions were defined as cultivated lands and a 1 km buffer zone surrounding settlements and roads, based on previous studies (Baloch et al. 2017; Naha et al. 2019) and actual field investigations. We overlaid the suitable habitats onto the human-dominated regions to estimate the extent and spatial distribution of overlap.

Results

The dispersal trend of Asian elephants

In the early 1990s, Herd I, originating from Xishuangbanna, initially dispersed to western Pu'er and periodically moved between Pu'er and Xishuangbanna (W1) (Fig. 2). Herd II dispersed to central Pu'er and established a resident herd in 1995 (M1). During the 2000s, Herd II continued its northward dispersal and settled in 2006 (M2). Concurrently, Herd III dispersed to the border between Pu'er and Xishuangbanna and formed a resident herd (M3). Due to Jinghong hydropower station construction, herd I separated, forming an independent resident population (W1), which proceeded westward after 2007 (W2, W3). The majority of the Asian elephant dispersal occurred during the 2010s. In 2010, Herd IV dispersed and formed the third resident herd within the middle region (M4). In 2011, Herd V established a new dispersal route heading northwards, and by the following year, it formed another resident herd (E2). Later, Herd V eventually returned back after several northward dispersals in the late 2010s (E3, E4). By 2020, the northward-moving herd had, for the first

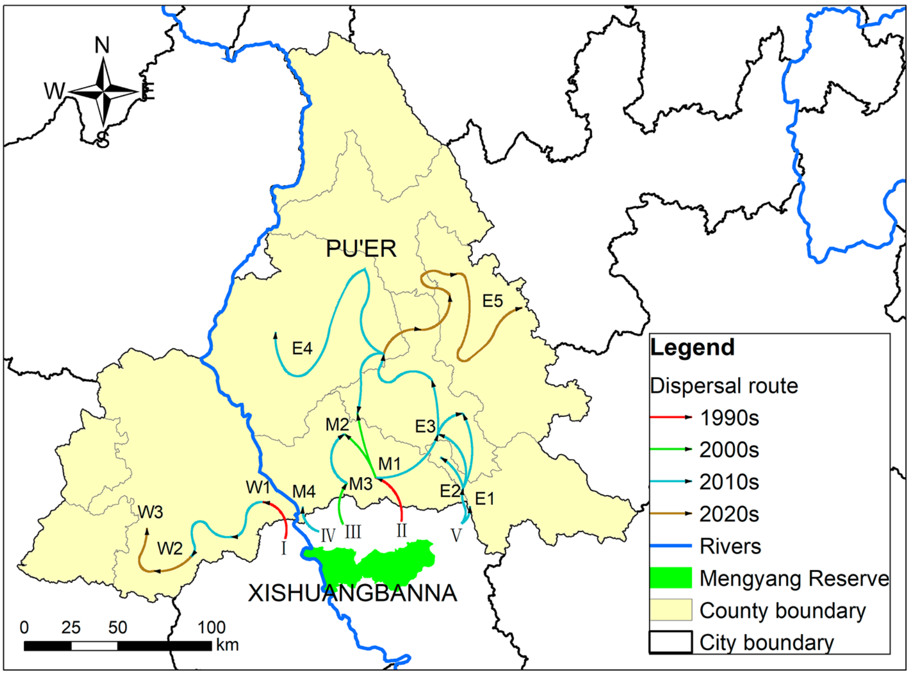

Fig. 2 The historical dispersal of the Asian elephants in Pu'er (**W**: the western part of Pu'er, **M**: the central part of Pu'er, **E**: the eastern part of Pu'er; **I-V**: the herds of the Pu'er-Banna Asian elephant population)

time, reached northeast Pu'er (E5). The dispersal process indicates that the Asian elephants, initially active in Xishuangbanna, shaped their current distribution over several decades of exploration and dispersal (Fig. 2).

The cumulative area of the 5 km buffer zone along the Asian elephant dispersal route experienced a remarkable increase from 719.37 km^2 to 9,730.02 km^2 (1352.6%) over the past 30 years. Over the same period, cultivated land increased from 54.28 km^2 to 1,074.42 km^2 (1979.4%) (Appendix Fig. 1). The expansion of the dispersal region in four periods (the 1990s, 2000s, 2010s, and 2020s) demonstrates an expanding distribution range with increasing dispersal distance that encompasses larger areas of the human-dominated regions represented by cultivated lands. Our findings indicate a gradual acceleration in the dispersal process of Asian elephants, particularly with a surge in dispersal events since 2010. Furthermore, compared to previous decades, the area of the dispersal region increased by 6.44 times (Appendix Fig. 1).

Asian elephant habitat under climate change

The average training AUCs for the replicate runs of past, present, and future periods in the MaxEnt model with bioclimatic variables were 0.930, 0.934, and 0.930 (with standard deviations of 0.001), respectively. The average discrepancy between the AUC values of the training set and the test set was 0.003, 0.004, and 0.003, respectively, indicating no overfitting occurred in the model.

The MaxEnt model statistical area was 131,742 km^2 (based on the grids). Figure 3 illustrates the distribution of suitable, sub-suitable, and unsuitable habitats across past, present, and future periods, while Table 1 provides corresponding area measurements and proportions.

The rank sum tests of pairwise combinations of climate suitability in the three periods revealed a statistically significant difference in climate between the three periods ($P < 0.01$). Compared to the present distribution range, 64.41% of the suitable habitat and 31.95% of the sub-suitable habitat remain stable over three periods. From the past period to the present period, the sub-suitable habitat of Asian elephants shrunk and gradually transformed into unsuitable habitats. Both suitable and sub-suitable habitats will expand slightly in the future (Table 1). Overall, the distribution of Asian elephant habitat has remained stable (Fig. 3).

Habitat selection of the Asian elephants

The average training AUCs for the replicate runs in the habitat MaxEnt model at three spatial scales with the environment and anthropogenic interference variables were 0.878, 0.960, and 0.940, with standard deviations of 0.002, 0.006, and 0.011, respectively. No overfitting occurred in the model, as evidenced by the average discrepancy between the AUC values of the training set and the test set – 0.008, 0.046, and 0.044. Table 2 presents the selected variables in the MaxEnt model across three spatial scales.

Based on the response curves of the critical variables, Asian elephants exhibited different habitat selection strategies across varying spatial scales. The suitable habitat at a 1 km scale was characterized by elevation between 800 and 1,200 m, slope under five degrees, and NDVI above 0.82. At the 5 km scale, the suitable habitat was characterized by elevation between 800 and 1,200 m, slope ranging from 1 to 5 degrees, road density less than 0.26,

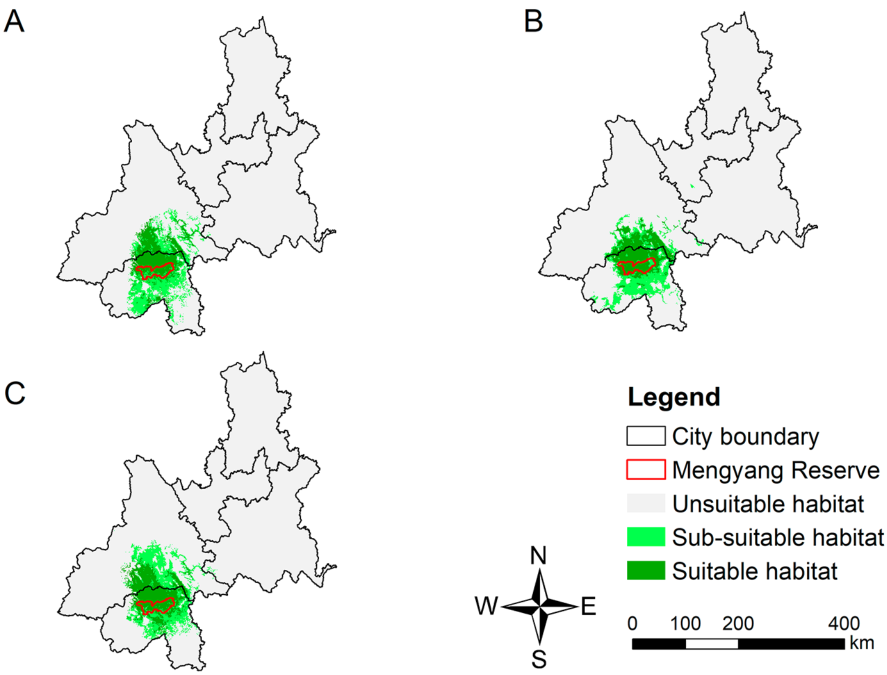

Fig. 3 Distribution of Asian elephant distribution range based on climate conditions in the past (**A**), present (**B**), and future (**C**)

Table 1 Area of Asian elephant habitat divided by climate in the past, present, and future periods

Period	Suitable habitat(km^2)	Sub-suitable habitat(km^2)	Unsuitable habitat(km^2)
Past	6,408(4.87%)	7,420(5.63%)	117,914(89.50%)
Present	6,468(4.91%)	6,503(4.94%)	118,771(90.15%)
Future	7,025(5.33%)	7,535(5.72%)	117,182(88.95%)

with south slope, southwest slope, and west slope as the predominant aspects, and NDVI above 0.9. At a larger spatial scale, 10 km, the suitable habitat was characterized by elevation between 700 and 1,200 m, road density less than 0.26, and settlement density below 0.07, broad-leaf forests, human modification less than 0.22, with south slope, southwest slope, and west slope as the predominant aspects, and forest-oriented land use type. The impact of anthropogenic factors on the habitat selection of Asian elephants was more significant at a larger spatial scale (Table 2).

The habitat distribution of Asian elephants

Of the study area calculated in the habitat MaxEnt model, 128,248 km^2, the suitable habitat spanned an area of 19,215 km^2 (14.98%), the sub-suitable habitat 25,121 km^2 (19.59%), and the unsuitable habitat 83,912 km^2 (65.43%). The regions of Xishuangbanna and southern Pu'er, which were the original habitats for Asian elephants and exhibited extensive connectivity among habitats, predominantly contained suitable and sub-suitable habitats. Addition-

Table 2 Relative contributions of selected variables at the 1, 5, and 10 km scales

1 km scale		5 km scale		10 km scale	
Variables	Contribution	Variables	Contribution	Variables	Contribution
Elevation	**71.4%**	**Elevation**	**50.6%**	**Elevation**	**21.6%**
Slope	**10.6%**	**Slope**	**8.8%**	**Road density**	**14%**
NDVI	**7.6%**	**Road density**	**7.5%**	**Settlement density**	**13.7%**
Vegetation	4.7%	**Aspect**	**6.8%**	**Vegetation**	**12.5%**
Human modification	2.7%	**NDVI**	**6%**	**Human modification**	**10.6%**
Land use	1.2%	Human modification	4.7%	**Aspect**	**7.3%**
Slop	1%	River density	3.4%	**Land use**	**5.4%**
Distance to road	0.8%	Vegetation	3.3%	Slope	4.7%
		Settlement density	3.3%	Patch density	4.5%
		Patch density	2.2%	River density	4.3%
		Distance to settlement	1.4%	NDVI	1.5%
		Land use	1.1%		
		Distance to road	1%		

*The critical variables (contribution > 5%) are indicated in bold

ally, southeastern Xishuangbanna, central and northwestern Pu'er, and areas between the Wuliang Mountains and Ailao Mountains were identified as suitable habitats. Only fragmented patches of suitable habitats were observed in the remaining study area (Fig. 4).

The current distribution status of the Pu'er-Banna population habitat indicated that the existing habitat surpasses any other potential habitats regarding suitability, size, and integrity (Fig. 4). We observed a lack of connectivity between the existing Asian elephant habitat and potential habitats.

Overlaying the distribution of Asian elephant habitats with layers related to human activities revealed that cultivated land accounted for 31.87% of the study area and 24.96% of the Asian elephants' distribution region (Fig. 5). Numerous human settlements and major roads were dispersed throughout the suitable habitat (Table 3). Within our study area, approximately 64.71% (28,689.26 km^2) of both suitable and sub-suitable habitats overlapped human-dominated regions; notably, within this overlapping region, around 60.98% (17,493.41 km^2) was within their distribution range (Table 3).

Discussion

In this study, we investigated the dispersal process of Asian elephants by tracing their historical movement, identified the distribution and suitability of their habitats, and analyzed the habitat selection strategy as influenced by human activities at various spatial scales. Our findings revealed a significant overlap between the habitats of Asian elephants and the human-dominated regions, indicating a firm reliance on and utilization of these regions by Asian elephants despite frequent human interactions. This dependence on human-dominated

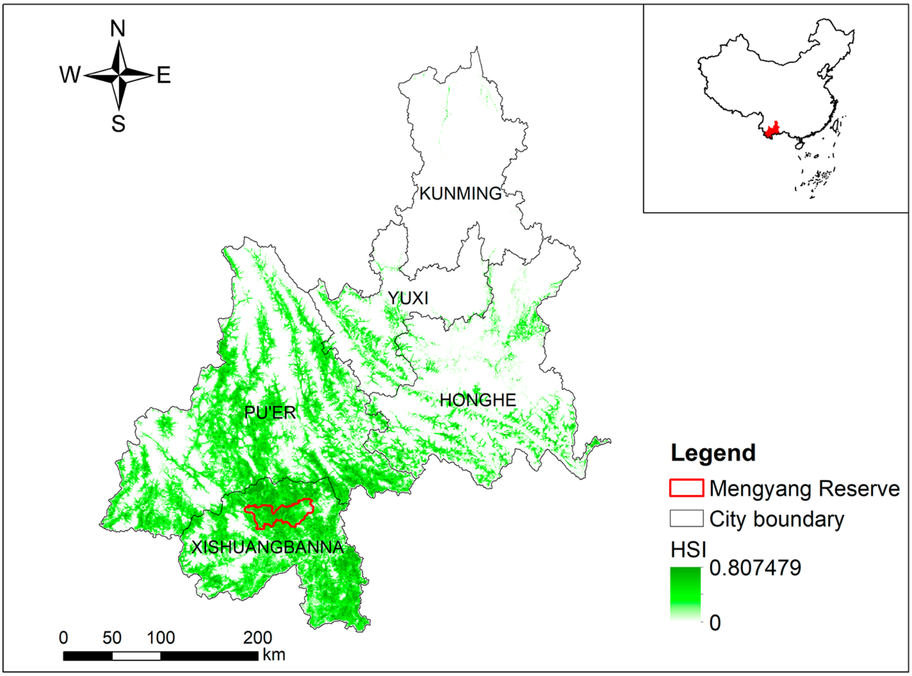

Fig. 4 Current habitat suitability of Asian elephants in the study area (HSI: Habitat Suitability Index)

regions can be attributed to the benefits of forest-agriculture matrices that are increasingly prevalent in such landscapes (Leimgruber et al. 2003; Tang et al. 2020).

Dispersal of Asian elephants

The dispersal of Asian elephants has been on the rise in recent decades, driven by various factors (Wang et al. 2017b; Zhu et al. 2019). As a category I national key protected wild animals in China, the population of Asian elephants has exhibited a positive growth trend in recent years (Li et al. 2009d; Zhang et al. 2006a; Yunnan Provincial Forestry and Grassland Administration, 2019, unpublished data). This growth can be attributed to the implementing of policies prohibiting hunting, logging, and burning practices, which accelerated vegetation recovery and enhanced forest canopy connectivity within the Mengyang reserve (Campos-Arceiz et al. 2021; Liu et al. 2020a). Although definitive identification of dominant factors influencing Asian elephant dispersal is lacking from available data sources, population growth and increased forest canopy connectivity significantly contributed to this process. The dispersal of the Asian elephant in the 2020s exhibited a lower magnitude compared to previous periods due to the limited temporal extent and the dominance of northward-moving herds. Based on long-term observations of dispersal patterns within the Pu’er-Banna Asian elephant population, the population will continue to disperse, as the human-dominated region and the Asian elephants’ distribution range expands gradually.

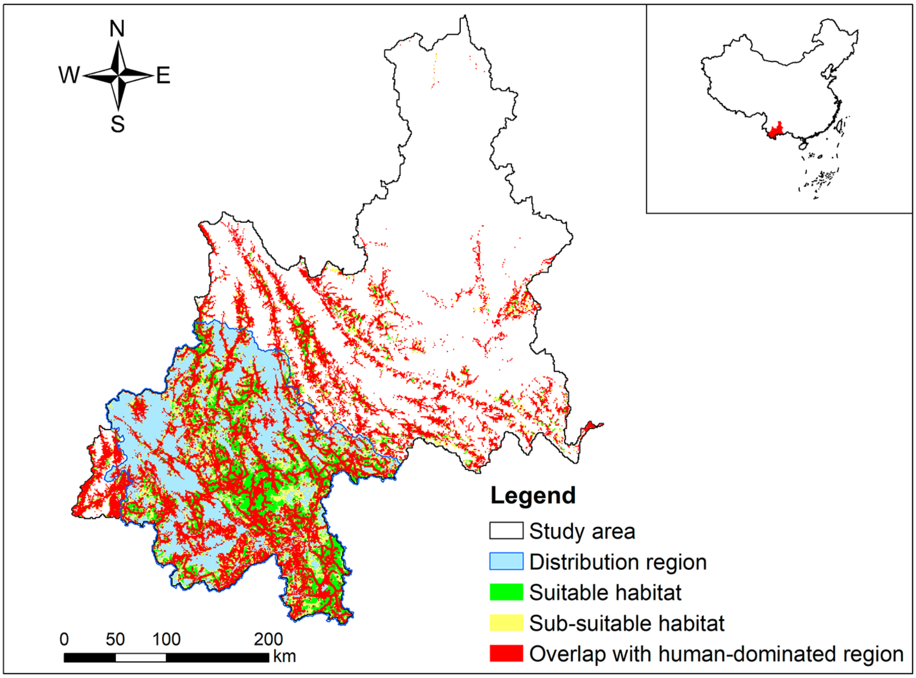

Fig. 5 The overlap between Asian elephant habitats and the human-dominated region in the study area

Table 3 The area of human-dominated regions within the Asian elephant habitat

Range	Suitable habitat	Sub-suitable habitat
Study area (km^2)	19,215	25,121
Road (km)	6,466.20	10,203.94
Settlement	1,760	2,895
Cultivated Land (km^2)	5,570.74	7,267.29
Overlap with human-dominated regions (km^2)	11,966.66	16,722.60

Previous research suggests that the dispersal direction is anticipated to be predominantly northward (Hickling et al. 2005; Moritz et al. 2008; Chen et al. 2011a; Li et al. 2015b).

Changes in the habitat distribution of Asian elephants under climate change

The direct impacts of global climate change can result in alterations to habitat range, fragmentation, and even gradual loss (Garcia-Valdes et al. 2015; Kanagaraj et al. 2019). Synergistic effects of habitat change, agricultural expansion, overexploitation by human activities, and land-use change can be important drivers of species extinction (Deb et al. 2019). The study suggests that the habitat distribution of Asian elephants remains predominantly stable in response to temporal changes in climate. The progressive conversion of habitats from sub-suitable to unsuitable during recent decades corresponds with the contraction of the Asian elephants' range. The gradual expansion of the suitable habitat for Asian elephants signifies their adaptive capacity to cope with ongoing climate change impacts. Even under the most severe climate change scenario (RCP8.5), projections indicate continued enlargement of the Asian elephant's habitat range.

Habitat selection of Asian elephants

The extensive loss of native habitats for Asian elephants is predominantly ascribed to human encroachment through urbanization and agricultural expansion (Chen et al. 2022c; Mariati et al. 2014; Sarma et al. 2008). Consequently, Asian elephants shift into human-dominated habitats and alter habitat selection strategies (Baskaran et al. 2011; Gubbi et al. 2014; Yu et al. 2015). Our findings indicate that the natural environment and anthropogenic interference variables significantly impact the habitat selection of Asian elephants. Although the relative importance of these variables varies, topographic features dominate across multiple spatial scales (Table 2). Gentle slopes or flat areas characterize the majority of suitable habitats for Asian elephants. Asian elephants avoid steep slopes and occasionally adopt "Z"-shaped routes to conserve energy while traversing such terrain (Sach et al. 2019). Furthermore, our study reveals that Asian elephants prefer habitats located at elevations between 700 and 1,200 m, despite their distribution range spanned 500-2,200 m (Appendix Figs. 2, 3 and 4).

The habitat selection of wild animals is determined by their specific requirements for space, food, water, and shelter (Yarrow 2009). These aspects also reflect the changes in habitat selection strategies of Asian elephants in human-dominated habitats.

The optimal foraging theory primarily reflects the influence of food on habitat selection: animals tend to acquire the highest food energy with the shortest feeding time (Pyke et al. 1977). The limited availability of palatable wild food plants and the allure of crops that offer accessibility, high nutritional value, and palatability incentivize Asian elephants to utilize high-risk human-dominated regions, although their habitat usage is based on a risk minimization strategy (Graham et al. 2009; Sivasubramanian and Ramakrishnan 2021).

Previous studies have demonstrated that the distribution and availability of water significantly influence the habitat selection of elephants (Shannon et al. 2006; Taher et al. 2021). We also found that distance to rivers and river density have minimal contribution to Asian elephants' habitat selection strategies (Table 2). It may be due to the abundance of rivers in our study area, which renders water sources less limiting for Asian elephants. Similarly, the low contribution of distance variables could be attributed to the dense distribution of rivers, roads, and settlements.

Asian elephants prefer broad-leaf forests with higher NDVI, encompassing primary, secondary, and plantation forests, which serve as essential habitats providing shelter and shade (Zhang 2007a; Huang et al. 2019; de la Torre et al. 2022). These forests typically exist within human-dominated landscapes as agroforestry ecotones comprising interspersed forest patches and cropland or adjacent secondary forests. Such habitats effectively fulfill their requirements for shelter and food (Nyhus and Tilson 2004; Huang et al. 2019). The presence of agroforestry zones further highlights the suitability of arable land as a habitat for Asian elephants.

Additionally, the impact of human activities on Asian elephant habitat selection can be inferred from the contribution of anthropogenic interference variables, which exhibit significant variations across different spatial scales (Table 2). At the small spatial scale, the influence of human activities is not readily discernible (cumulative contribution of anthropogenic interference variables 4.7%). At the medium spatial scale, human activities primarily manifest through road distribution (cumulative contribution of anthropogenic interference variables 18%). At a larger spatial scale, variables such as human modification, land use, and distribution of roads and settlements significantly impact Asian elephants' habitat selection (cumulative contribution of anthropogenic interference variables 43.7%) (Table 2). The findings demonstrate that the spatial distribution of human-dominated regions and the intensity of human activities exert an increasingly significant influence on Asian elephant habitat selection at a larger scale, potentially serving as determinant indicators for assessing habitat suitability. Furthermore, our research reveals that although lower levels of human modification, roads, and settlement densities are more favorable for Asian elephants, their dependence on human-dominated regions diminishes the likelihood of selecting regions without human intervention. It is plausible that Asian elephants are undergoing adaptation processes and developing dependent on human-dominated regions.

The overlap between suitable habitat of Asian elephants and human-dominated regions

We found a substantial degree (64.71%) of spatial overlap between suitable and sub-suitable habitats for Asian elephants and human-dominated regions. While anthropogenic interference and land-use change pose limitations on Asian elephant habitats (Sampson et al. 2019; Tripathy et al. 2021), linear infrastructures such as high-grade roadways also impede Asian elephant movements (Blake et al. 2008; Dasgupta and Ghosh 2015; Huang et al. 2020). The coexistence of human activities and elephant habitats increases human-elephant conflicts (Wilson et al. 2015b). Despite these challenges, Asian elephants benefit from their habitat overlapping with human-dominated regions. The relatively flat terrain in these regions facilitates the movement of Asian elephants, while cultivated lands provide accessibility to food resources. Agroforestry ecotone becomes their ideal habitat in human-dominated regions (Huang et al. 2020). Moreover, China's wildlife protection policy effectively safeguards Asian elephants against direct hunting threats. The lower human population density in Southwest China allows Asian elephants to tolerate relatively low levels of anthropogenic interference (Xu 2017). These conditions contribute significantly to the extensive overlap between Asian elephant habitats and human-dominated regions. Based on this, regions of human-elephant overlap should prioritize the optimization of agricultural structures, establish sustainable food sources to support elephants and implement conservation measures to protect lives and property in conflict-prone regions. The reinforcement of monitoring and early warning systems and the guidance for the dispersal of Asian elephant populations away from settlements and major roads are imperative. This approach will effectively contribute to protecting and managing Asian elephants, mitigating conflicts with human activities.

Conclusions

Our research has demonstrated that Asian elephants are capable of adapting to climate change and maintaining habitat stability over time. We also revealed that human activities are increasingly influencing the distribution and selection of habitats for Asian elephants at a larger spatial scale. Recent studies have indicated that identifying the dominant factors driving habitat change is crucial to understanding behavioral changes in Asian elephants and mitigating human-elephant conflicts (Bai et al. 2022; Campos-Arceiz et al. 2021; Wang et al. 2021c). Our study also revealed a high overlap between Asian elephant habitats and human-dominated regions, which is unlikely to change in the near future, leading to changes in Asian elephant behavior, frequent human-elephant contact, difficulties in managing and protecting them, and threatening the lives and property of local residents (Liu et al. 2017b; Neupane et al. 2020; Tang et al. 2020; Wilson et al. 2015b). This finding highlights the need for effective management strategies to minimize the negative impacts of human activities on the movement and distribution of these animals and alleviate the human-elephant conflicts. In conclusion, our study provides important insights into the factors affecting the habitat selection strategy of Asian elephants. It highlights the potential risks of Asian elephants' increasing dependence on human-dominated regions. These findings have important implications for the conservation and management of Asian elephants.

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Declarations

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